

Sentiment Analysis of Product Reviews using Naïve Bayes

Vrishaan Taneja : 23FE10CSE00111
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1 Introduction of Problem Statement

In the present world, sentiment analysis has become a major part of day-to-day life, from home automation to analysing product reviews. One of the major uses of sentiment analysis is to classify a review as negative or positive. Many solutions have been developed to classify product reviews into the two categories. Large and complex models like BERT are able to perform these tasks really well, but the major disadvantage of these models is the resources they require.

Therefore, many lighter approaches to Natural Language Processing (NLP) techniques are used to automate sentiment classification with lower overhead.

The objective of this project is to build a machine learning model using TF-IDF feature extraction and the Multinomial Naïve Bayes algorithm. Additionally, utilising pre-processing techniques such as cleaning, lemmatization, are compared to choose the best model.

2 Contribution of the Work

The key contributions are:

1. Implementation of multiple preprocessing techniques, including cleaning, lemmatization, and stemming.
2. Development of a sentiment classification model using TF-IDF and Naïve Bayes.
3. Evaluation using confusion matrix, ROC curve, and classification metrics.

3 Dataset Description and Visualisation

The dataset consists of product reviews labelled as positive and negative.

a total of 400k (four hundred thousand) reviews were taken from a Kaggle dataset (amazon)

Column	Description
Label	Sentiment (1 = Negative, 2 = Positive)
Title	Review title
Review	Review content
Text	Combined text

Table 1: Dataset Description

The dataset is preprocessed by removing null values, cleaning text, and splitting into training and testing sets (80:20).

4 Feature Extraction

TF-IDF is used for feature extraction.

$$TF = \frac{\text{Term Frequency}}{\text{Total Terms}}, \quad IDF = \log \frac{N}{df}$$

$$TF-IDF = TF \times IDF$$

5 Algorithms Used

5.1 1. Multinomial Naïve Bayes (Implemented)

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

5.2 2. Logistic Regression (Theoretical)

$$P(y = 1|x) = \frac{1}{1 + e^{-z}}$$

5.3 3. Support Vector Machine (Theoretical)

$$w \cdot x + b = 0$$

6 Hyperparameters and Training

Parameter	Value
Vectorizer	TF-IDF
N-gram	(1,2)
Model	MultinomialNB
Split	80:20

Table 2: Hyperparameters

7 Experimental Results

7.1 Confusion Matrix

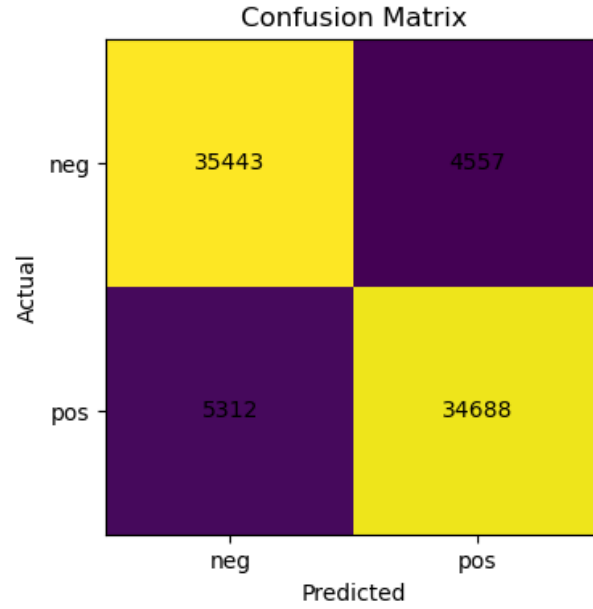


Figure 1: Confusion Matrix

7.2 ROC Curve

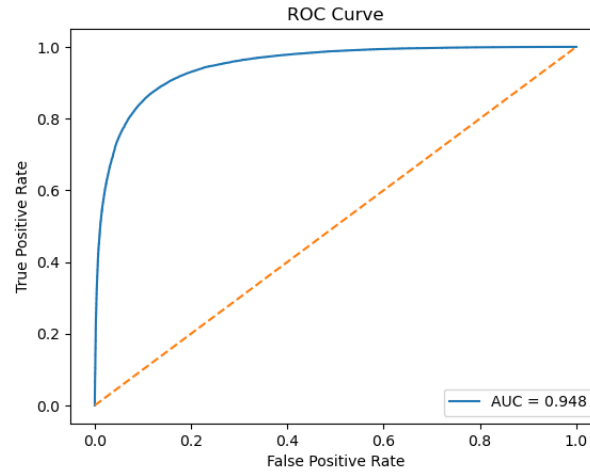


Figure 2: ROC Curve

7.3 Evaluation Metrics

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2PR}{P + R}$$

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===== CLEAN =====
Accuracy: 0.8766375
Confusion Matrix:
[[35443  4557]
 [ 5312 34688]]
Classification Report:
              precision    recall  f1-score   support

      neg       0.870      0.886      0.878      40000
      pos       0.884      0.867      0.875      40000

   accuracy                   0.877      80000
  macro avg       0.877      0.877      0.877      80000
 weighted avg       0.877      0.877      0.877      80000


===== LEMMA =====
Accuracy: 0.87295
Confusion Matrix:
[[35396  4604]
 [ 5560 34440]]
Classification Report:
              precision    recall  f1-score   support

      neg       0.864      0.885      0.874      40000
      pos       0.882      0.861      0.871      40000

   accuracy                   0.873      80000
  macro avg       0.873      0.873      0.873      80000
 weighted avg       0.873      0.873      0.873      80000


===== STEM =====
Accuracy: 0.8751375
Confusion Matrix:
[[35789  4211]
 [ 5778 34222]]
Classification Report:
              precision    recall  f1-score   support

      neg       0.861      0.895      0.878      40000
      pos       0.890      0.856      0.873      40000

   accuracy                   0.875      80000
  macro avg       0.876      0.875      0.875      80000
 weighted avg       0.876      0.875      0.875      80000


{'clean': 0.8766375, 'lemma': 0.87295, 'stem': 0.8751375}

```

Figure 3: Evalutaion Metrics

8 Conclusion

In this report, a sentiment analysis model was successfully developed using TF-IDF feature extraction and the multinomial Naive Bayes algorithm.

Techniques such as cleaning and lemmatization were found to slightly influence the model performance. In particular, the Clean dataset provided a better score when compared to lemmatization and stemming.

The evaluation metrics used were accuracy, recall, precision, and F1 score to choose the best model and a confusion matrix and ROC curve on the best selected model to verify if the model was well-suited for the dataset.

It may be observed that there are still a few instances being classified wrongly, as the dataset might contain sentences that contain sarcasm.

9 References

- Han, J., Kamber, M. (2012)
- Olson, D., Delen, D. (2008)
- Jurman, G. et al. (2012)